TEEGALA KRISHNA REDDY ENGINEERING COLLEGE

(Approved By AICTE , Affiliated To JNTUH, Accredited By NBA And NAAC)



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ABSTRACT

This project proposes a smart digital platform that connects farmers, agricultural workers, and landowners on a single network, similar to a professional linking system for agriculture. The system provides AI-based worker recommendations using location, skills, availability, ratings, and cost. A wage prediction module helps ensure fair pricing for both farmers and workers. Crop and season-based labor planning simplifies workforce estimation. Voice-based interaction with Telugu support improves accessibility for rural users. Additional features such as worker availability calendars, emergency labour requests, land-sharing options, and weather alerts enhance decision-making. The platform aims to improve efficiency, transparency, and trust in agricultural labour and land management.

INTRODUCTION

Agriculture is still the driving force behind the Indian economy, but farmers still have issues with accessing reliable labour, determining a fair wage rate, and effectively utilizing land resources. Most of the current agricultural applications available are related to advisory, equipment rental, or market prices, but labour management is still done manually by networking with people. This results in time wastage, disputes over wages, and unavailability of labour during peak agricultural seasons. With the rise of smartphone usage in rural areas, there is a great need for a platform that brings farmers, agricultural labour, and landowners together in an organized and trustworthy manner. This proposal brings forth a smart-agri workforce application that utilizes AI recommendations, voice interaction, and availability tracking to make labour and land sharing easier.

EXISTING SYSTEM

In the existing system, agricultural labor management is mostly carried out through manual methods such as personal contacts, local labor contractors, and word-of-mouth communication. While several agricultural mobile applications are available, they primarily focus on advisory services, farm machinery rental, and market price information rather than labor hiring and land management. Applications like FARMS – Farm Machinery Solutions, Kisan Sarathi, Farmitra, and KhetEase support farmers with guidance and information but do not provide a structured digital system for workforce availability, wage transparency, or land-sharing. As a result, farmers face operational inefficiencies, especially during peak agricultural seasons.

Drawback or Limitations

- Finding workers takes more time
- No clear idea of worker availability
- Wage rates are not transparent
- Frequent misunderstanding and disputes
- Not useful during urgent labour needs

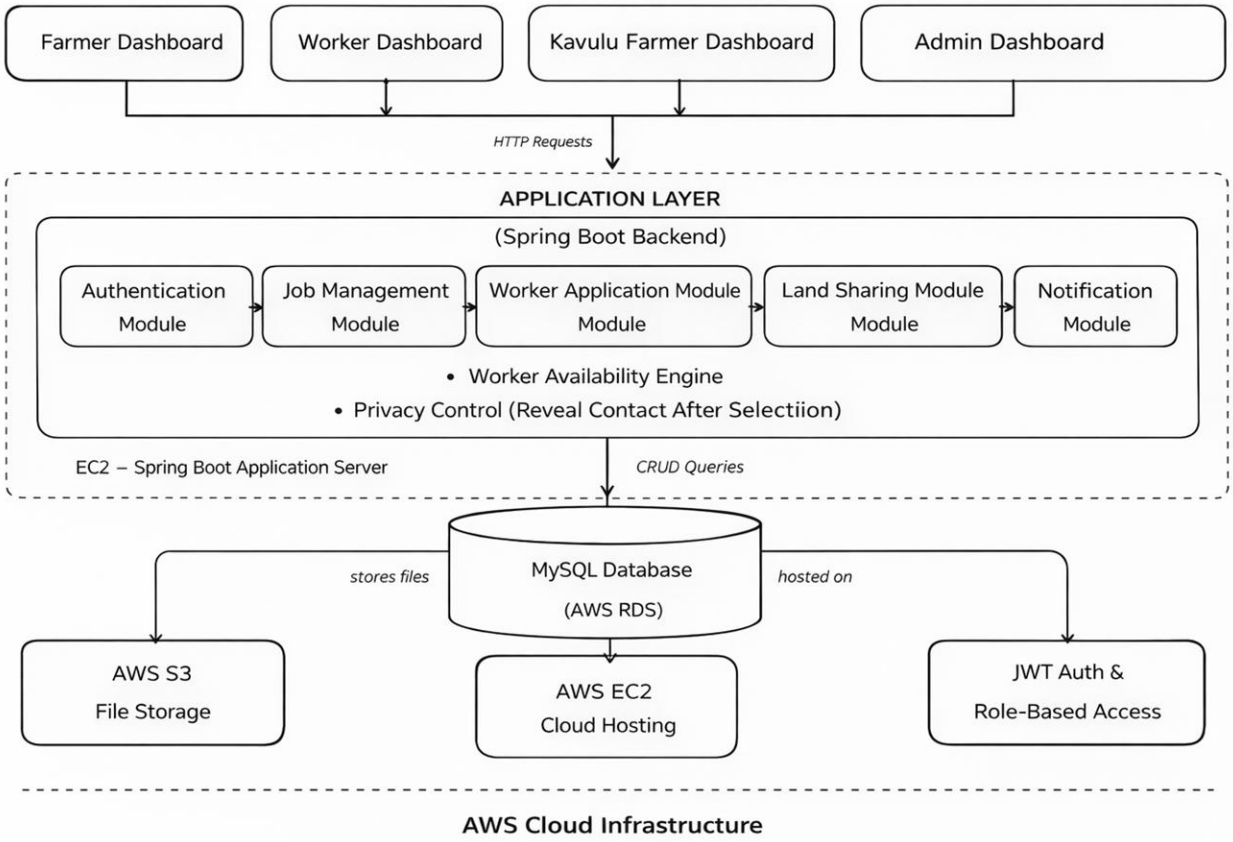
PROPOSED SYSTEM

The proposed system is a smart digital platform that connects farmers, agricultural workers, and landowners through a single application. It uses AI-based recommendations to suggest suitable workers based on location, skills, availability, and ratings. The system also provides wage estimation, voice-based interaction, and real-time worker availability tracking. Additional features such as emergency labour requests, land-sharing options, and weather alerts help farmers make better decisions. Overall, the platform simplifies labour hiring and land management while improving transparency and efficiency.

Advantages of Proposed System

- Easy and quick worker selection
- Fair and transparent wage estimation
- Voice support for rural users
- Real-time worker availability tracking
- Emergency labour support
- Better use of land resources

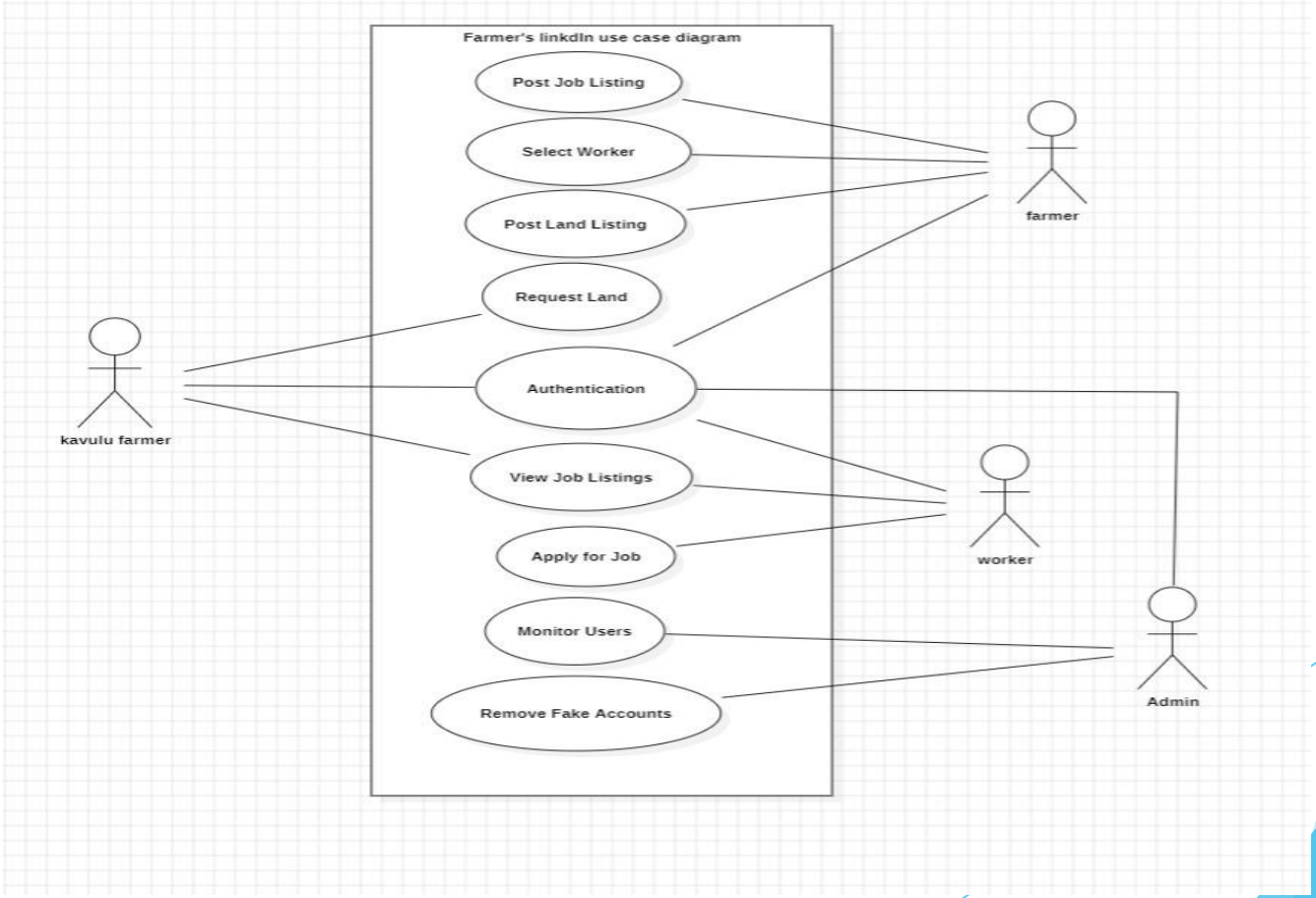
System Architecture



UML Diagrams

Use Case Diagram

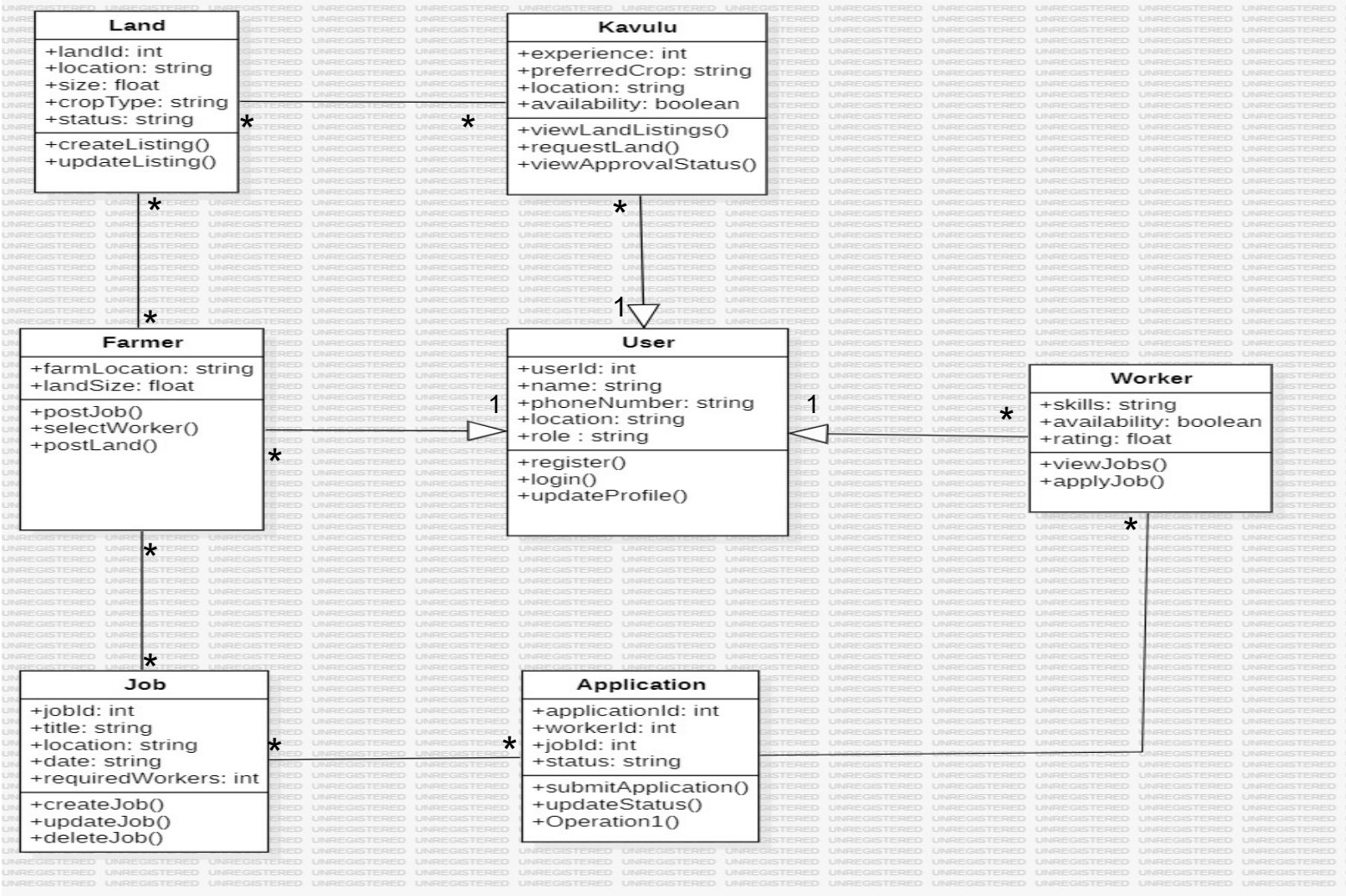
Shows who uses the Farmer LinkedIn system (**Farmer, Worker, Kavulu-Farmer, Admin**) and what actions they can perform such as posting jobs, applying for jobs, requesting land, and managing the platform.



UML Diagrams

Class Diagram

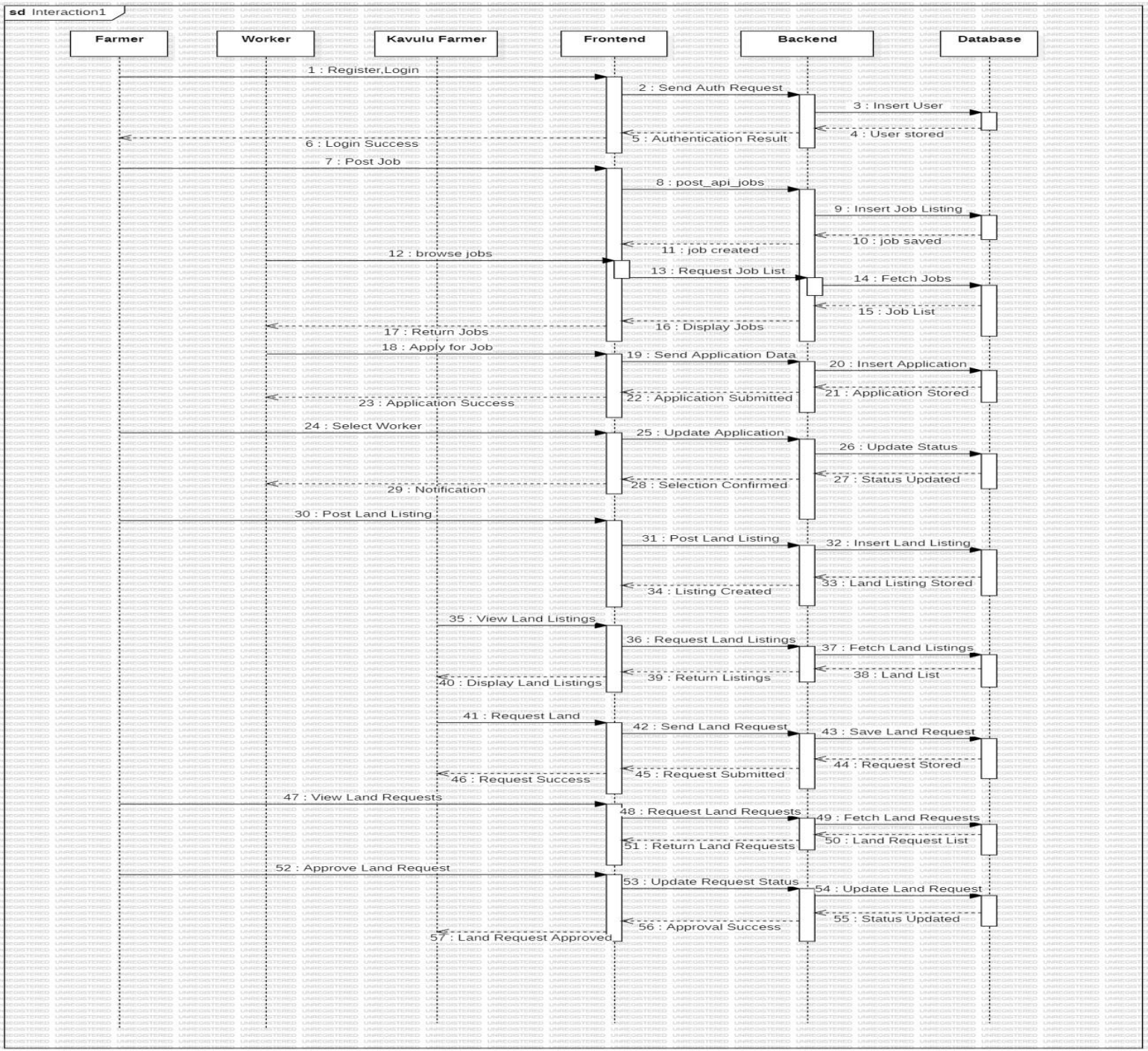
Shows the structure of the Farmer LinkedIn system by representing the main classes (User, Job, Application, LandListing, Notification, Admin) and the relationships between them.



UML Diagrams

Sequence Diagram

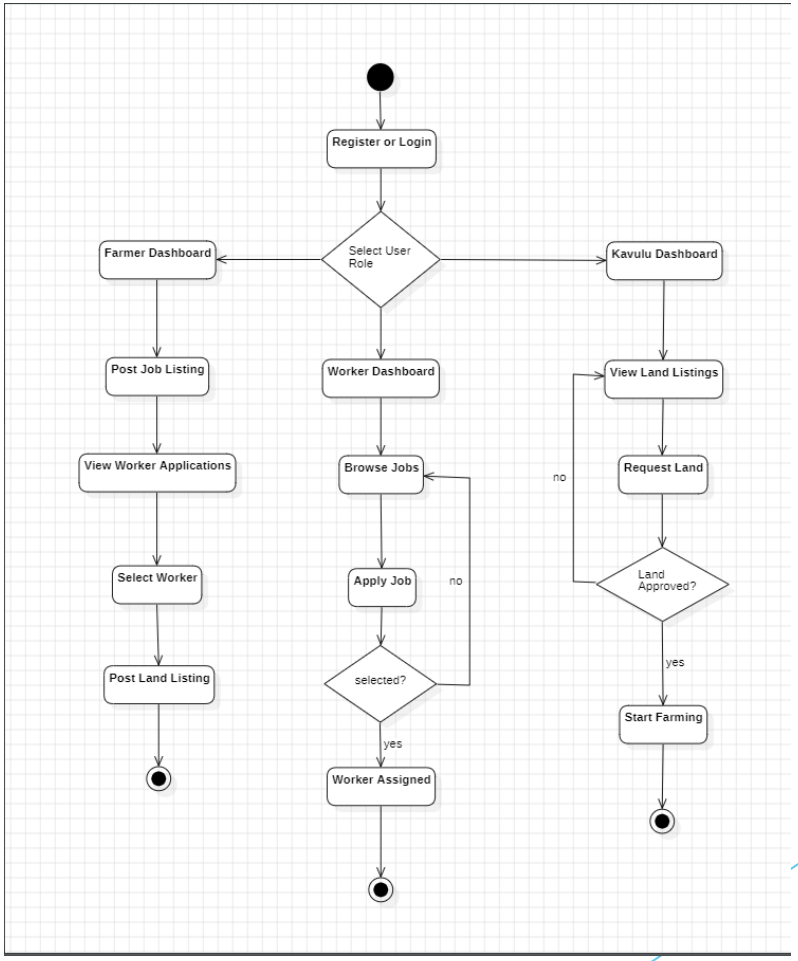
Shows the order of interactions between different components (Farmer, Worker, System/Backend, Database) over time to complete a process such as posting a job, applying for a job, and selecting a worker in the Farmer LinkedIn system.


sd Interaction1
Farmer
Worker
Kavulu Farmer
Frontend
Backend
Database
1 : Register,Login
2 : Send Auth Request
3 : Insert User
4 : User stored
5 : Authentication Result
6 : Login Success
7 : Post Job
8 : post_api_jobs
9 : Insert Job Listing
10 : job saved
11 : job created
12 : browse jobs
13 : Request Job List
14 : Fetch Jobs
15 : Job List
16 : Display Jobs
17 : Return Jobs
18 : Apply for Job
19 : Send Application Data
20 : Insert Application
21 : Application Stored
22 : Application Submitted
23 : Application Success
24 : Select Worker
25 : Update Application
26 : Update Status
27 : Status Updated
28 : Selection Confirmed
29 : Notification
30 : Post Land Listing
31 : Post Land Listing
32 : Insert Land Listing
33 : Land Listing Stored
34 : Listing Created
35 : View Land Listings
36 : Request Land Listings
37 : Fetch Land Listings
38 : Land List
39 : Return Listings
40 : Display Land Listings
41 : Request Land
42 : Send Land Request
43 : Save Land Request
44 : Request Stored
45 : Request Submitted
46 : Request Success
47 : View Land Requests
48 : Request Land Requests
49 : Fetch Land Requests
50 : Land Request List
51 : Return Land Requests
52 : Approve Land Request
53 : Update Request Status
54 : Update Land Request
55 : Status Updated
56 : Approval Success
57 : Land Request Approved

UML Diagrams

Activity Diagram:

Shows the workflow of the AgroConnect system by illustrating the sequence of activities such as user registration, posting jobs, applying for jobs, selecting workers, and requesting land, along with the decision points in the process.



System Specifications

1. **Hardware Requirements**

Processor: Intel i5 or equivalent
RAM: Minimum 4 GB (8 GB recommended)
Storage: 256 GB SSD or higher
Internet Connection: Required for cloud deployment

2. **Software Requirements**

Operating System: Windows
Programming Language: Java
Framework: Spring Boot
Frontend Technologies: HTML, CSS, JavaScript
Database: MySQL
Cloud Platform: AWS (EC2, RDS, S3)
Development Tools: VS Code
Web Browser: Google Chrome

3. Technologies Used

Backend: Spring Boot (Java)
Database Management: MySQL (AWS RDS)
Cloud Hosting: AWS EC2
File Storage: AWS S3
API Communication: REST APIs
Security: JWT Authentication and Role-Based Access Control

MODULES

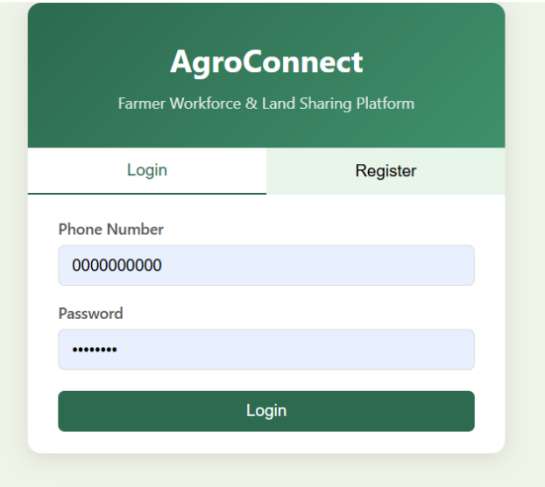
1. User Authentication Module

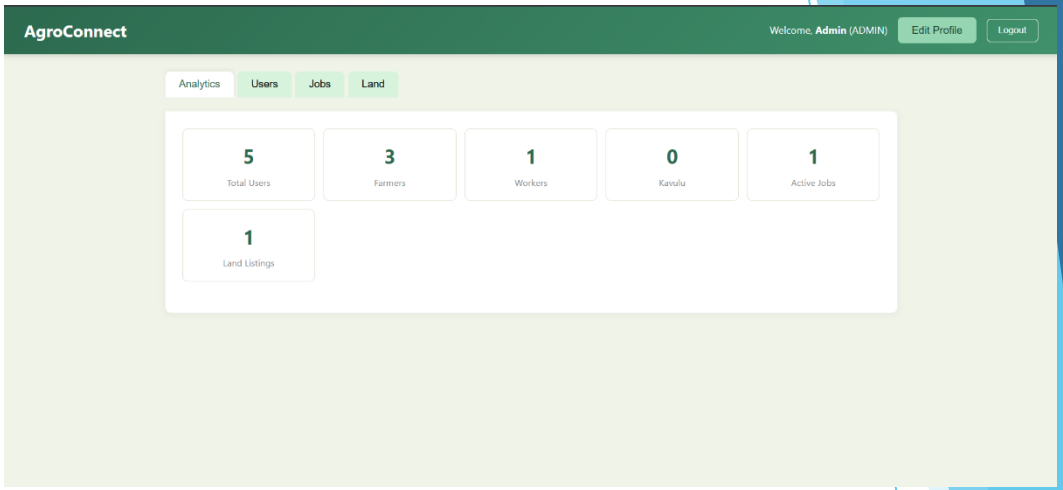
2. Job Management Module

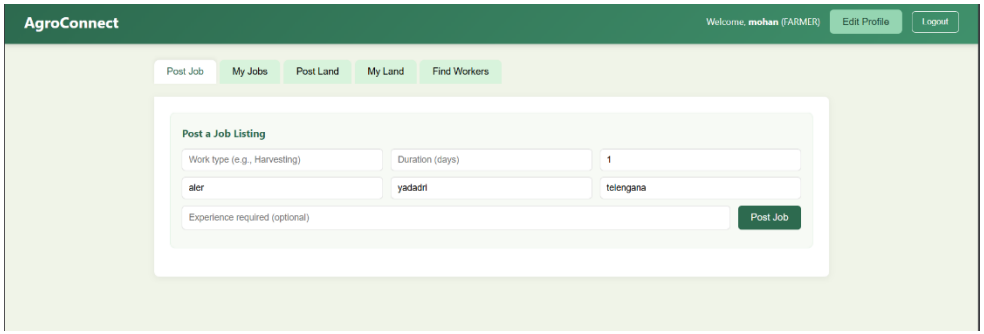
3. Land Management Module

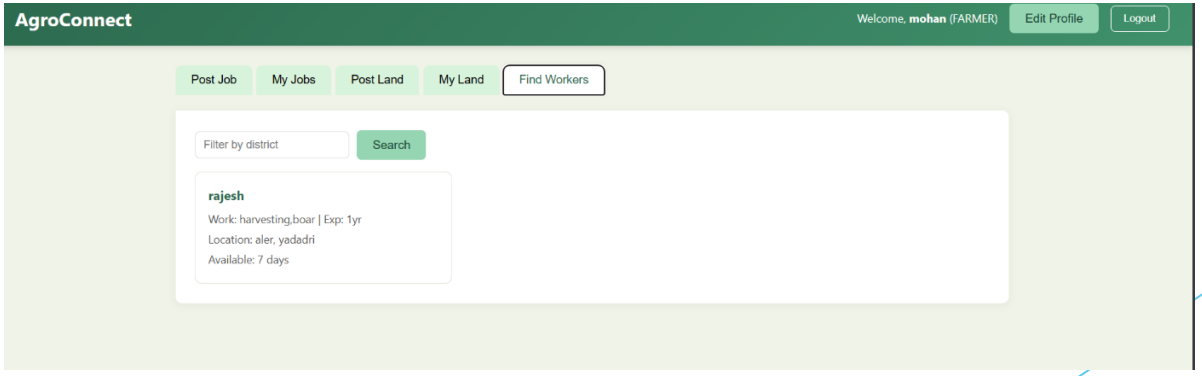
4. Admin Management Module

Output Screens









Conclusion

AgroConnect successfully provides a digital platform that connects farmers, workers, and kavulu farmers in a simple and efficient way . The system enables farmers to post job requirements and land availability, while workers and kavulu farmers can easily find and apply for opportunities based on their needs and location . By eliminating communication gaps and improving accessibility, AgroConnect enhances coordination in agricultural activities . The project demonstrates how technology can be used to simplify real-world problems in the farming sector . Futureimprovements can include payment integration, advanced location tracking, and mobile application support.

References

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UML Official Documentation – https://www.uml.org/
W3Schools (Web Development Reference) – https://www.w3schools.com/
Stack Overflow (Developer Community) – https://stackoverflow.com/
GitHub Documentation – https://docs.github.com/

THANK YOU